Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

Centre Number | Candidate Number

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Half Past Paper – Week 28

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1	Solving single inequalities	2	U759
3	Sharing amounts in a given ratio, Finding fractions and percentages of amounts	5	U577, U881, U554
5	Angles in polygons	3	U427
7	Volume of cuboids, Calculating with density, Writing and simplifying ratios	3	U786, U910, U687
9	Finding the surface area of cubes and cuboids	3	U929
11	Calculating with ratios and algebra	3	U676
13	Using Pythagoras' theorem in 2D, Finding the area of circles	3	U385, U950
15a	Geometric proofs with vectors	2	U560
15b	Solving geometric problems using vectors	3	U781
17	Constructing direct proportion equations, Constructing inverse proportion equations	4	U407, U138
19	Adding and subtracting algebraic fractions, Using the quadratic formula	4	U685, U665
21	Finding the area of sectors	5	U373
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Solve $7x - 27 < 8$

.....

(Total for Question 1 is 2 marks)

3 A delivery company has a total of 160 cars and vans.

the number of cars : the number of vans = 3 : 7

Each car and each van uses electricity or diesel or petrol.

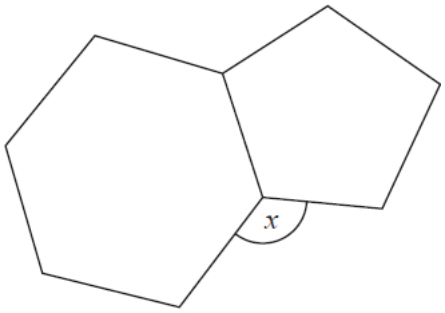
$\frac{1}{8}$ of the cars use electricity.
25% of the cars use diesel.
The rest of the cars use petrol.

Work out the number of cars that use petrol.
You must show all your working.

.....

(Total for Question 3 is 5 marks)

5 Here is a regular hexagon and a regular pentagon.

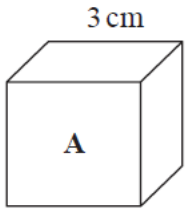


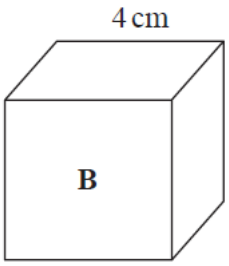
Work out the size of the angle marked x.
You must show all your working.

.....°

(Total for Question 5 is 3 marks)

7 Here are two cubes, **A** and **B**.





Cube **A** has a mass of 81 g.

Cube **B** has a mass of 128 g.

Work out

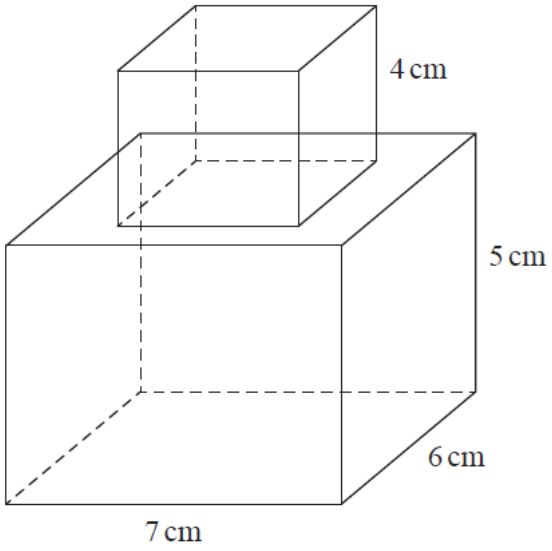
the density of cube **A** : the density of cube **B**

Give your answer in the form $a : b$, where a and b are integers.

.....

(Total for Question 7 is 3 marks)

9 A cube is placed on top of a cuboid, as shown in the diagram, to form a solid.



The cube has edges of length 4 cm.
The cuboid has dimensions 7 cm by 6 cm by 5 cm.

Work out the total surface area of the solid.

..... cm^2

(Total for Question 9 is 3 marks)

11 Cormac has some sweets in a bag.
The sweets are lime flavoured or strawberry flavoured or orange flavoured.

In the bag

$$\begin{array}{c}\text{number of lime}\\ \text{flavoured sweets}\end{array} : \begin{array}{c}\text{number of strawberry}\\ \text{flavoured sweets}\end{array} : \begin{array}{c}\text{number of orange}\\ \text{flavoured sweets}\end{array} = 9 : 4 : x$$

Cormac is going to take at random a sweet from the bag.

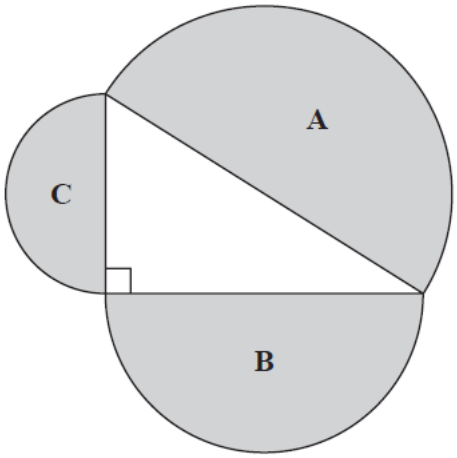
The probability that he takes a lime flavoured sweet is $\frac{3}{7}$

Work out the value of x.

$x =$ .....

(Total for Question 11 is 3 marks)

13 A right-angled triangle is formed by the diameters of three semicircular regions, **A**, **B** and **C** as shown in the diagram.



Show that

area of region **A** = area of region **B** + area of region **C**

(Total for Question 13 is 3 marks)

15 A, B and C are three points such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b}$$
$$\overrightarrow{AC} = 15\mathbf{a} + 20\mathbf{b}$$

(a) Prove that A, B and C lie on a straight line.

(2)

D, E and F are three points on a straight line such that

$$\overrightarrow{DE} = 3\mathbf{e} + 6\mathbf{f}$$
$$\overrightarrow{EF} = -10.5\mathbf{e} - 21\mathbf{f}$$

(b) Find the ratio

length of DF : length of DE

.....

(3)

(Total for Question 15 is 5 marks)

17 y is directly proportional to the square root of t.
$y = 15$ when $t = 9$

t is inversely proportional to the cube of x.
$t = 8$ when $x = 2$

Find a formula for y in terms of x.
Give your answer in its simplest form.

.....

(Total for Question 17 is 4 marks)

19 Solve $\frac{1}{2x-1}+\frac{3}{x-1}=1$

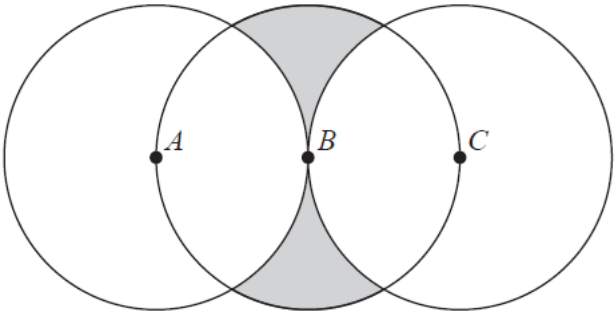
Give your answer in the form $\frac{p\pm\sqrt{q}}{2}$ where p and q are integers.

.....

(Total for Question 19 is 4 marks)

21 The diagram shows three circles, each of radius 4 cm.

The centres of the circles are A, B and C such that ABC is a straight line and $AB = BC = 4$ cm.



Work out the total area of the two shaded regions.
Give your answer in terms of π

..... cm^2

(Total for Question 21 is 5 marks)

TOTAL FOR PAPER IS 40 MARKS